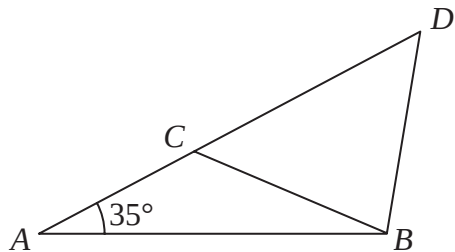


7.



Not to scale

Figure 3

Figure 3 shows the design for a structure used to support a roof.

The structure consists of four wooden beams, AB , BD , BC and AD .

Given $AB = 6.5$ m, $BC = BD = 4.7$ m and angle $BAC = 35^\circ$

(a) find, to one decimal place, the size of angle ACB ,

(3)

(b) find, to the nearest metre, the total length of wood required to make this structure.

(3)